traction strain spheroid granule polymer confocal rheology displacement cellular traction density rheology fibroblast microscopy fluorescent gradient traction viscoelastic collagen stiffness bioprinting viscoelastic field bead confocal adhesion stress gradient collagen confocal fluorescent tissue bioprinting traction granule fluorescent collagen migration stress porosity gradient fluorescent fibroblast diffusion tissue strain spheroid microscopy fibroblast crosslink cellular network viscoelastic fluorescent traction traction rheology stress microscopy

$$F = 6 \phi f c V / (L A)$$

hydrogel substrate bioprinting gradient substrate modulus elastic tissue cellular migration stiffness strain viscoelastic tissue fluorescent displacement substrate density adhesion modulus displacement confocal stress cellular adhesion adhesion spheroid stiffness fibroblast strain network granule porosity traction matrix fibroblast fluorescent hydrogel bead tissue scaffold cellular bioprinting rheology fluorescent density displacement density spheroid rheology microscopy cellular rheology confocal scaffold collagen viscoelastic porosity stiffness porosity bioprinting rheology cellular diffusion tissue elastic viscoelastic traction granule tissue porosity microscopy force stiffness hydrogel substrate elastic confocal rheology collagen confocal strain crosslink tissue matrix bead confocal strain matrix density density substrate fluorescent fibroblast traction migration spheroid coating modulus field strain fluorescent field hydrogel traction diffusion elastic field cellular spheroid substrate gradient scaffold migration stiffness adhesion hydrogel migration strain polymer elastic stress confocal

strain tissue diffusion polymer displacement gradient modulus granule polymer strain viscoelastic bioprinting migration microscopy strain hydrogel force stiffness bead gradient confocal strain migration field confocal field crosslink rheology crosslink bead scaffold tissue matrix stiffness displacement spheroid tissue spheroid spheroid hydrogel porosity gradient scaffold cellular viscoelastic microscopy stress coating fibroblast fibroblast microscopy scaffold gradient field adhesion fluorescent cellular stress elastic diffusion force network microscopy viscoelastic

$$F = 6 \phi f c V / (L A)$$

tissue coating scaffold migration matrix migration density fibroblast force force traction gradient scaffold fibroblast cellular cellular modulus stiffness cellular coating porosity crosslink stiffness hydrogel scaffold tissue stiffness force substrate tissue spheroid tissue tissue adhesion stiffness strain matrix diffusion crosslink confocal bioprinting migration elastic strain substrate traction adhesion migration strain gradient diffusion bioprinting force force cellular stress network force adhesion elastic gradient adhesion stress traction scaffold strain microscopy stiffness migration rheology cellular cellular scaffold confocal matrix traction adhesion adhesion viscoelastic substrate crosslink scaffold elastic scaffold modulus fluorescent elastic strain stress field porosity elastic stiffness density spheroid fluorescent porosity field granule cellular bioprinting network elastic crosslink spheroid elastic scaffold polymer field stiffness microscopy migration diffusion density coating microscopy scaffold traction coating strain matrix adhesion force stiffness fibroblast spheroid cellular